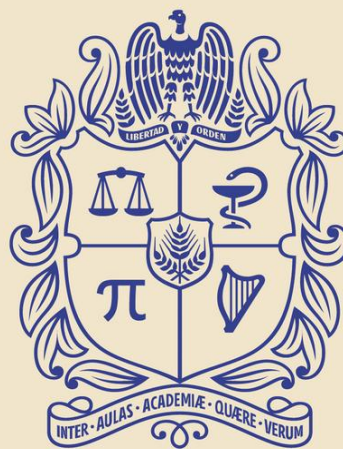




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ROOTS OF EQUATIONS

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1. Introduction



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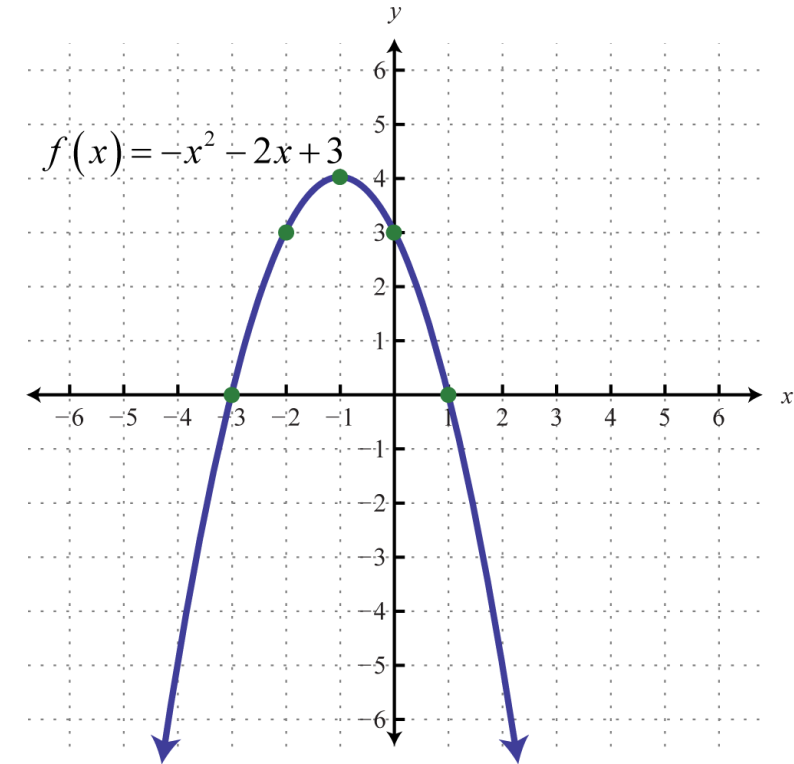
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Definition of a Root (Zero) of an Equation

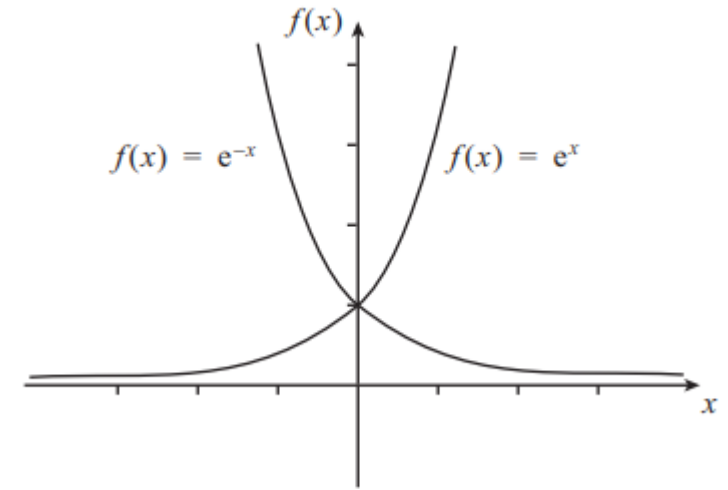
- A root is any value of x that makes $f(x)=0$, also called a “zero” of the equation.
- Example: the quadratic formula finds roots of $ax^2+bx+c=0$ explicitly.





Need for Numerical Methods

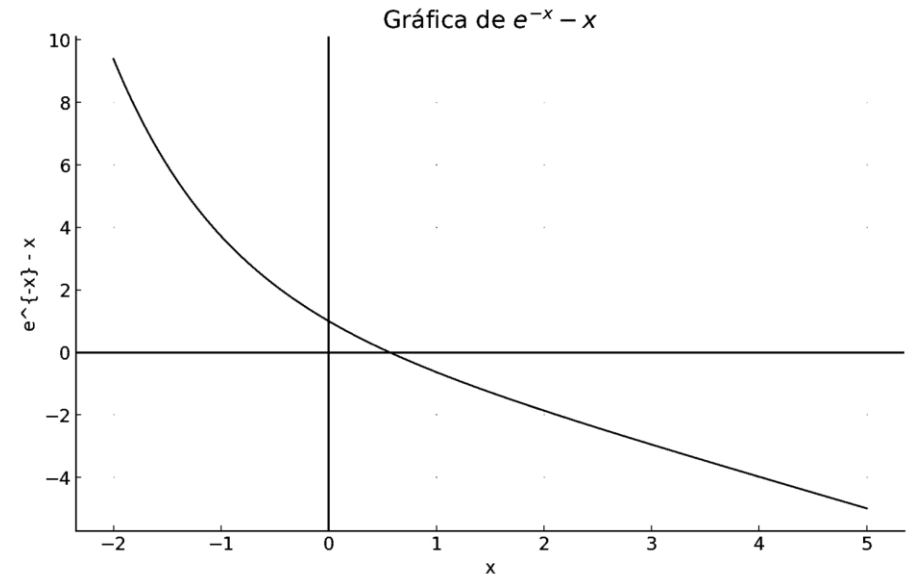
- Many functions (for example, $f(x) = e^{-x} - x$) lack closed-form solutions for roots.
- Numerical techniques enable approximate, systematic determination of roots when direct methods fail.





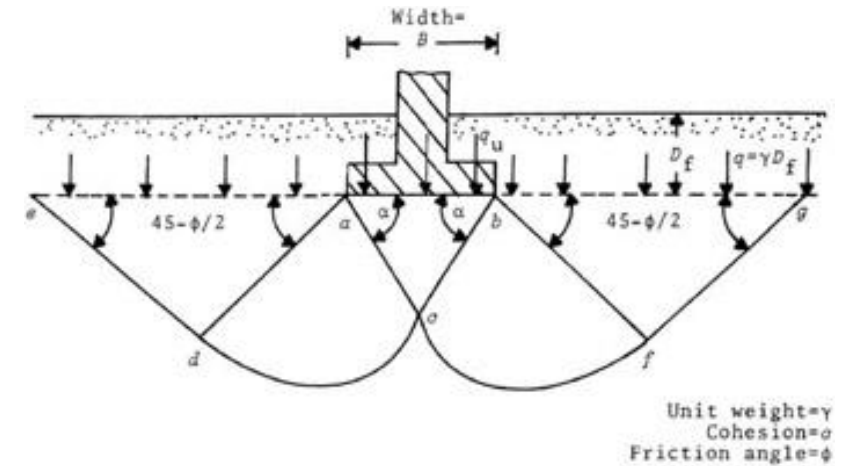
Non-Computer Approximate Techniques

- **Graphical Methods:** Plot $f(x)$ and eyeball the x-intercept for a rough estimate.
- **Trial and Error:** Guess an x , compute $f(x)$, then iterate toward $f(x) \approx 0$ —**inefficient and imprecise for engineering needs.**



Roots in Engineering Design

- Many design problems require solving for implicit parameters by finding the root of a rearranged equation $f(\text{parameter})=0$.
- Fundamental principles leading to root-finding models include heat, mass, force and energy balances; Newton's laws.





Roots in Engineering Design

Fundamental Principle	Dependent Variable	Independent Variable	Parameters
Heat balance	Temperature	Time and position	Thermal properties of material and geometry of system
Mass balance	Concentration or quantity of mass	Time and position	Chemical behavior of material, mass transfer coefficients, and geometry of system
Force balance	Magnitude and direction of forces	Time and position	Strength of material, structural properties, and geometry of system
Energy balance	Changes in the kinetic- and potential-energy states of the system	Time and position	Thermal properties, mass of material, and system geometry
Newton's laws of motion	Acceleration, velocity, or location	Time and position	Mass of material, system geometry, and dissipative parameters such as friction or drag
Kirchhoff's laws	Currents and voltages in electric circuits	Time	Electrical properties of systems such as resistance, capacitance, and inductance



Mathematical Background

- **Algebraic Equations:** Polynomial forms $a_0 + a_1x + \dots + a_nx^n = 0$.
- **Transcendental Equations:** Nonalgebraic forms (e.g., trigonometric, exponential, logarithmic).
- Roots can be real or complex; two main problem areas:
 - Locating a single real root (algebraic or transcendental).
 - Finding all real and complex roots of polynomials.



Organization

- **Bracketing Methods:** Bisection, false-position, incremental searches with guaranteed convergence.
- **Open Methods:** Fixed-point iteration, Newton–Raphson, secant, plus extension to nonlinear systems.
- **Polynomial Roots:** Müller’s method, and use of Excel/MATLAB for polynomials.
- **Engineering Case Studies:** Applications across disciplines, trade-off comparisons, and summary of key formulas.



2. Bracketing Methods



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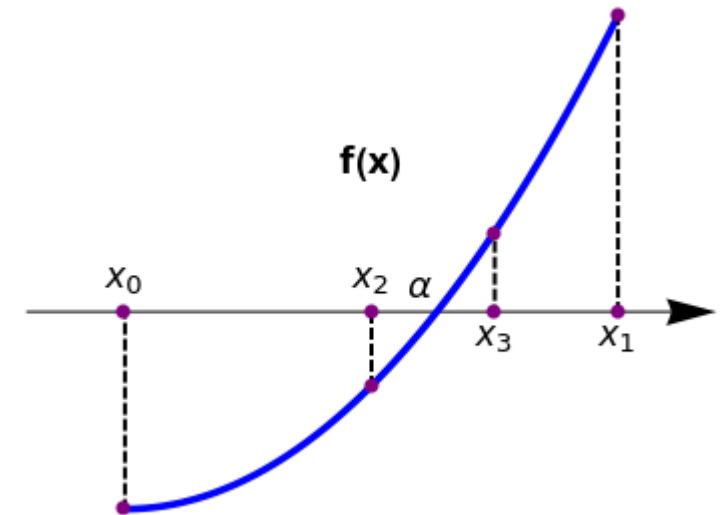
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Bracketing Methods Overview

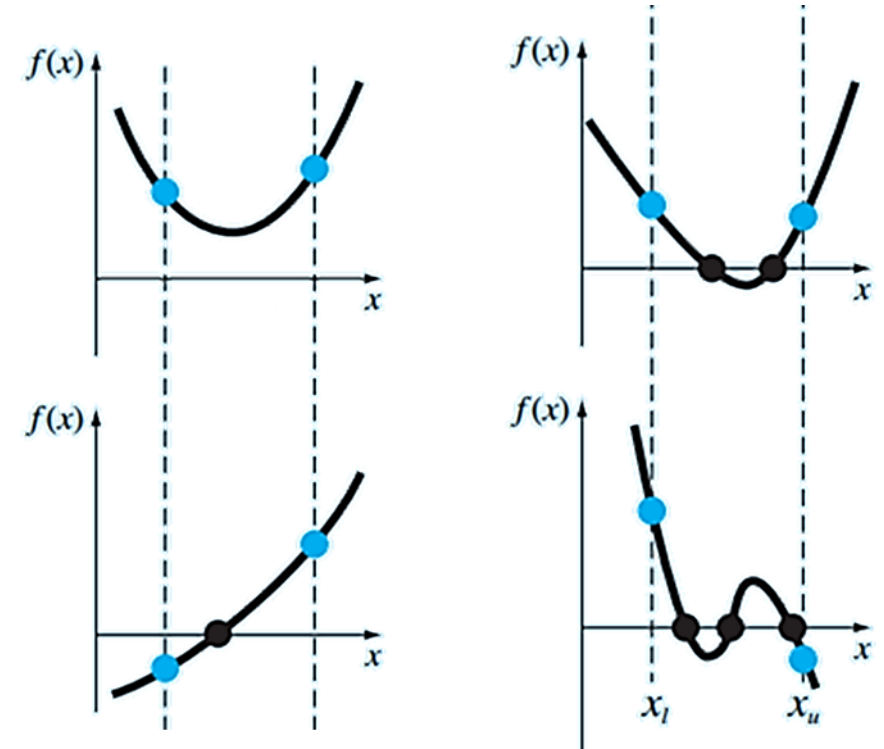
- Exploit the **sign change** of a continuous function $f(x)$ near a root to locate it.
- Require **two initial guesses**, x_l and x_u , such that $f(x_l)$ and $f(x_u)$ **have opposite signs**, thereby “bracketing” the root.
- Systematic strategies are then applied to **narrow the interval** $[x_l, x_u]$ until the root is isolated with desired accuracy.





Purpose of Graphical Methods

- Provide a **visual, rough estimate** of where $f(x) = 0$ by plotting the function and observing its x-axis crossings.
- The x-coordinate of each crossing yields an approximate root value, useful for initial guesses in more precise numerical methods.





Limitations of Graphical Techniques

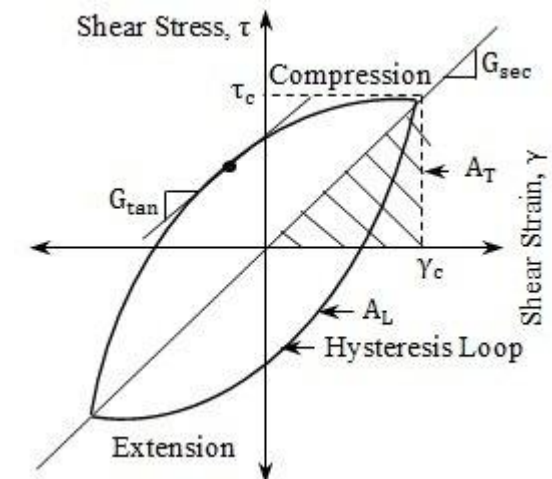
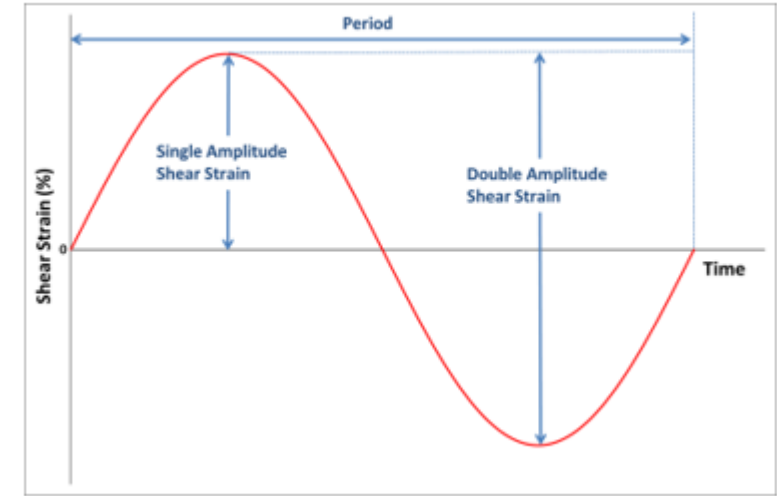
- **Lack precision:** Cannot deliver accurate root values on their own.
- Best employed to generate **starting points** for algorithms like bisection, false-position, or secant methods.





Additional Benefits of Graphical Interpretation

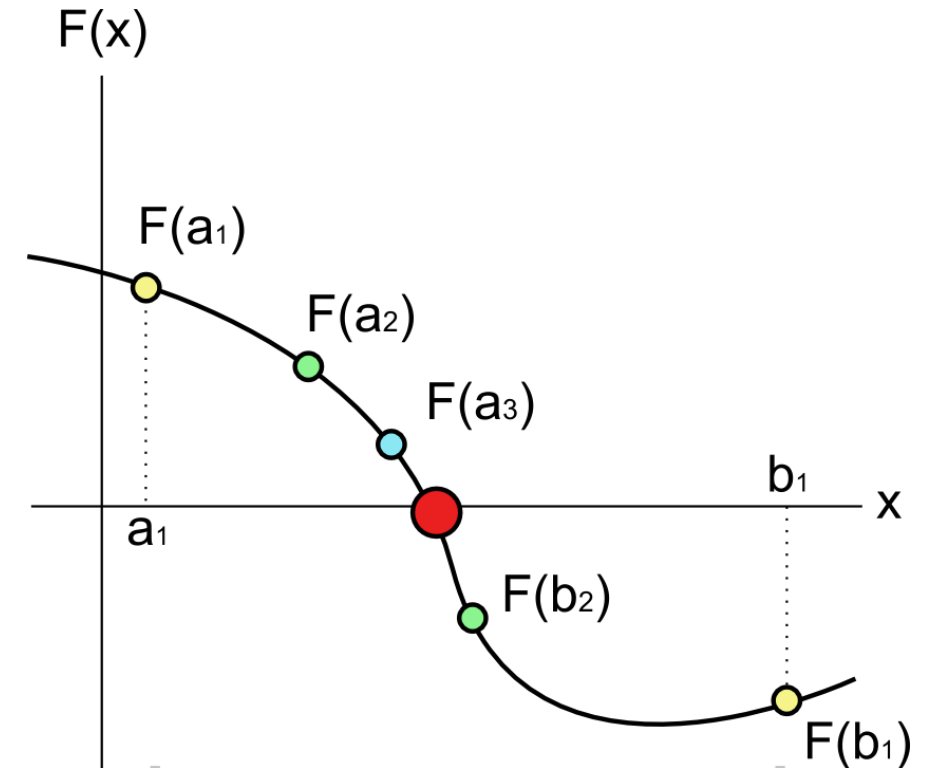
- Helps engineers **visualize function behavior** (e.g., monotonicity, oscillations).
- Aids in identifying potential **pitfalls** of numerical methods, such as regions of flat slope or rapid variation.





Interpreting Sign Changes and Root Multiplicity

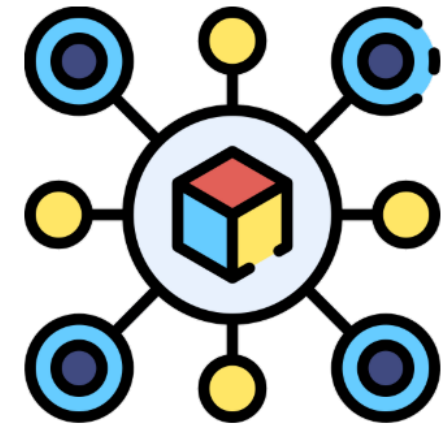
- If $f(x_l)$ and $f(x_u)$ have **opposite signs**, the Intermediate Value Theorem guarantees **an odd number of roots** within $[x_l, x_u]$.
- If they share the **same sign**, **there may be no roots** or **an even number of roots** in that interval.





Practical Implications for Engineers

- While graphical methods alone are not sufficient, when **combined** with bracketing strategies they become **powerful tools**.
- These techniques form the foundation for solving a wide range of **root-finding problems** encountered in engineering analyses and applied mathematical modeling.



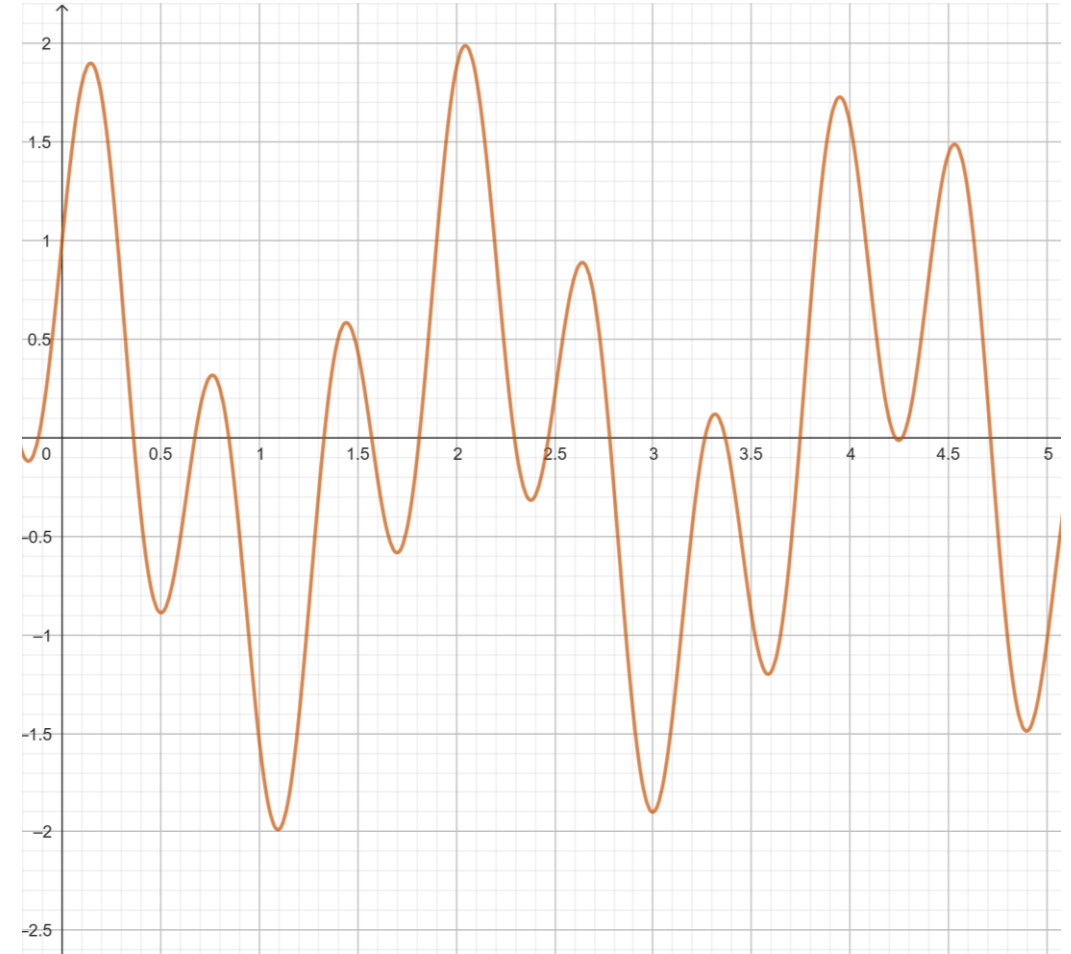


Example 1

Graphical plotting can greatly aid in uncovering the zeros of a function. Consider the function

$$f(x) = \sin(10x) + \cos(3x)$$

on the interval $0 \leq x \leq 5$. Produce a plot of $f(x)$ over this range to explore its behavior and estimate where it crosses the x -axis.





Example 1

1. Generate the x vector

```
x = linspace(0,5,1000);
```

- **`linspace(a,b,n)`** creates a row vector of **n** linearly spaced points between **a** and **b** .
- Here, x has 1000 evenly spaced values from 0 to 5, giving a smooth curve when plotted.

2. Compute the function values

```
y = sin(10*x) + cos(3*x);
```

- MATLAB applies $\sin(10*x)$ and $\cos(3*x)$ element-wise to the vector x .
- The result y is another 1×1000 vector, where each entry $y_i = \sin(10x_i) + \cos(3x_i)$.



Example 1

3. Plot the curve

`plot(x,y)`

- Draws a 2D line plot of y versus x.
- By default, MATLAB connects the points in order with a **solid blue line**.

4. Add a grid

`grid on`

- Turns on the background grid, making it easier to see where the curve crosses the axes.



Example 1

5. Label the axes

```
xlabel('x');  
ylabel('f(x)');
```

- `xlabel` places the text “x” below the horizontal axis.
- `ylabel` places the text “f(x)” beside the vertical axis, clarifying that the plotted values are $f(x)$.

6. Draw the zero line

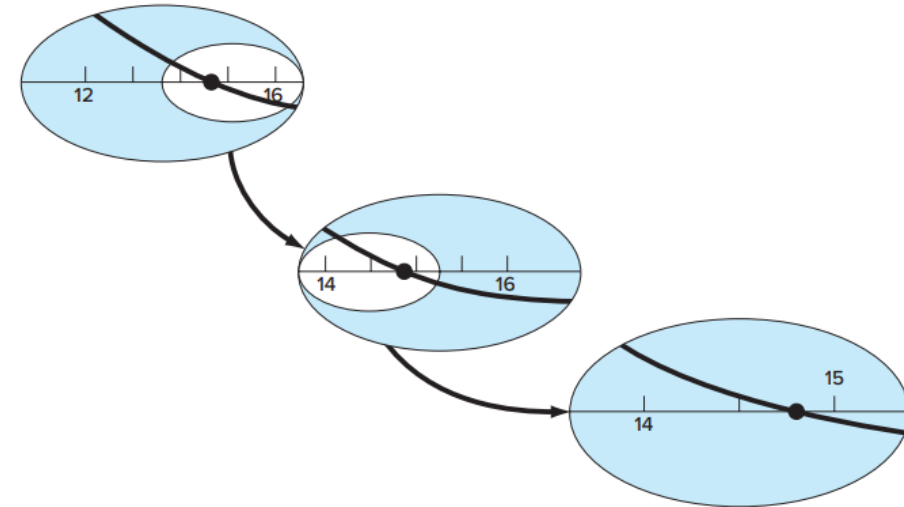
```
yline(0,'--')
```

- `yline(y0,LineStyle)` draws a horizontal line.
- Here, it’s a dashed ('--') line at $y=0$, so you can immediately see each point where the blue curve crosses that line—i.e., the roots.



Fundamental principle of the bisection method

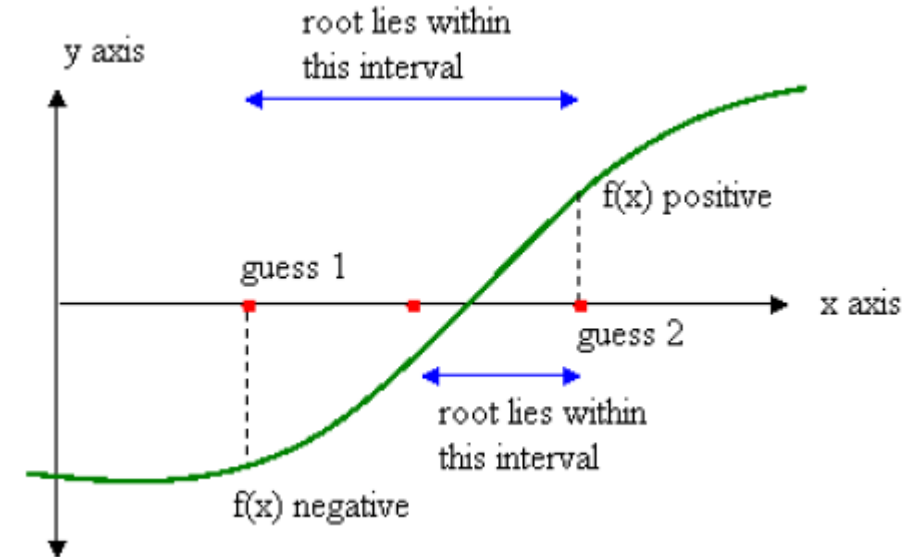
- Applicable when $f(x)$ is continuous on $[x_l, x_u]$ and $f(x_l) \cdot f(x_u) < 0$, guaranteeing at least one root in the interval.
- Also known as binary chopping, interval halving, or Bolzano's method.
- At each iteration, compute the midpoint $x_r = (x_l + x_u)/2$ and evaluate $f(x_r)$.
- Replace either x_l or x_u with x_r depending on the sign of $f(x_r)$, thereby halving the bracketing interval.





Incremental-search context

- General incremental searches locate a sign change by subdividing an interval into smaller parts.
- Bisection is a special case where each subinterval is exactly **half** the previous one.



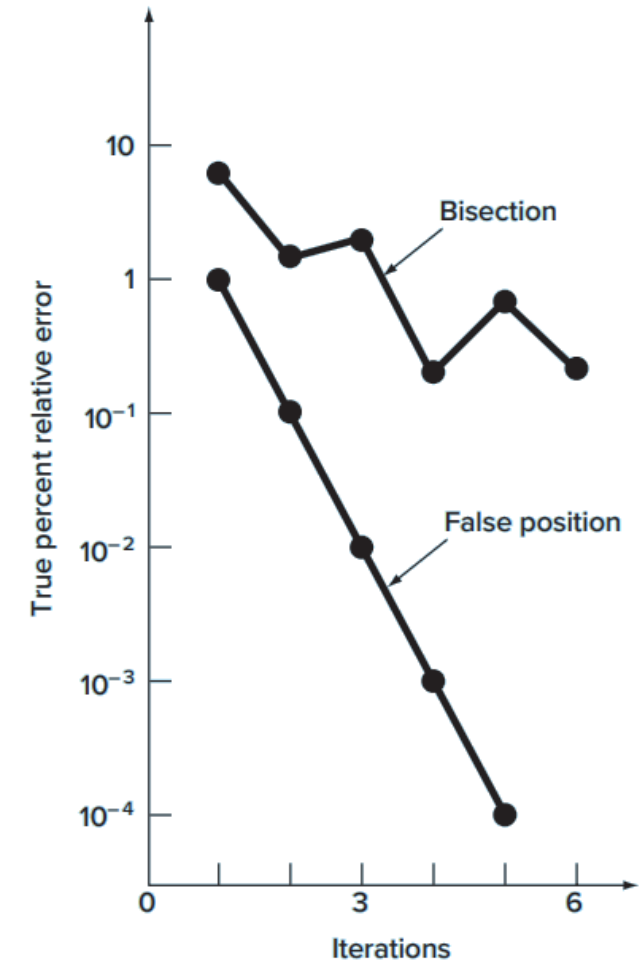
$$x_r = \frac{x_l + x_u}{2}$$

If $f(x_l) f(x_r) < 0$,
then $x_u := x_r$;
otherwise $x_l := x_r$



True error vs. approximate error behavior

- True error ε_t may not decrease monotonically with each iteration.
- However, the interval width Δx_i containing the root is halved every iteration, ensuring predictable error reduction.





Approximate percent relative error

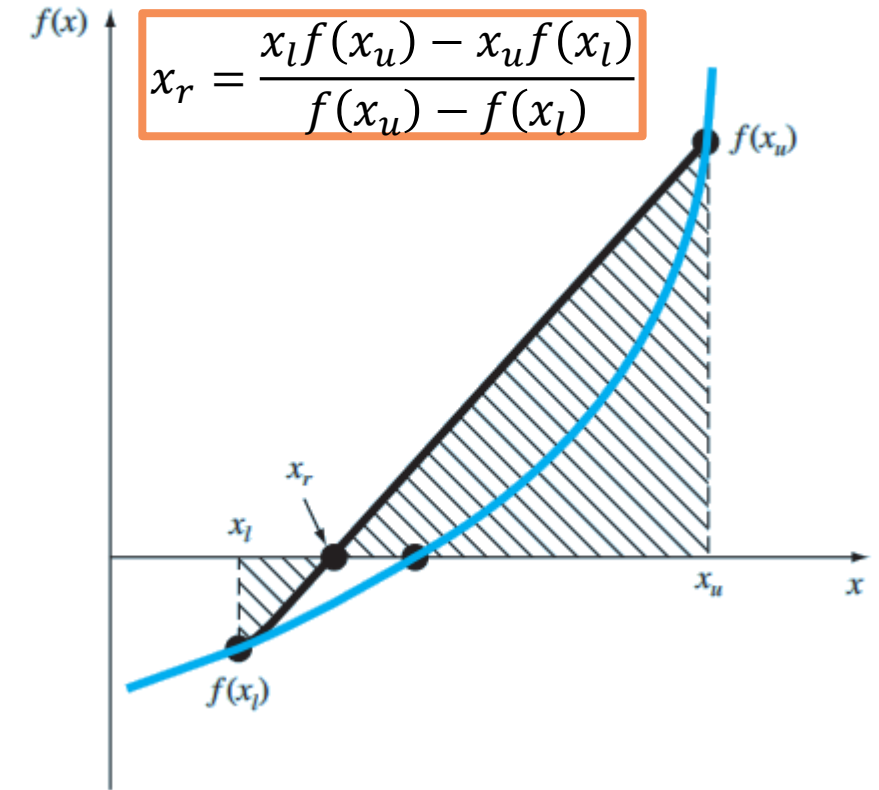
$$\varepsilon_a = \left| \frac{x_r^{new} - x_r^{old}}{x_r^{new}} \right| \times 100\%$$

- Uses only current and previous approximations, requiring no knowledge of the true root.
- Computation terminates when ε_a falls below a prescribed stopping criterion ε_s .



The False-Position Method

- Bisection splits x_l, x_u equally, **ignoring** relative magnitudes of $f(x_l)$ and $f(x_u)$.
- If $|f(x_l)| \ll |f(x_u)|$, the true root lies **closer to** x_l .
- False position draws a **straight line** between $(x_l, f(x_l))$ and $(x_u, f(x_u))$ and uses its intersection with the x-axis as an improved estimate.
- Also known as the **linear interpolation method**.



If $f(x_l) f(x_r) < 0$,
then $x_u := x_r$;
otherwise $x_l := x_r$

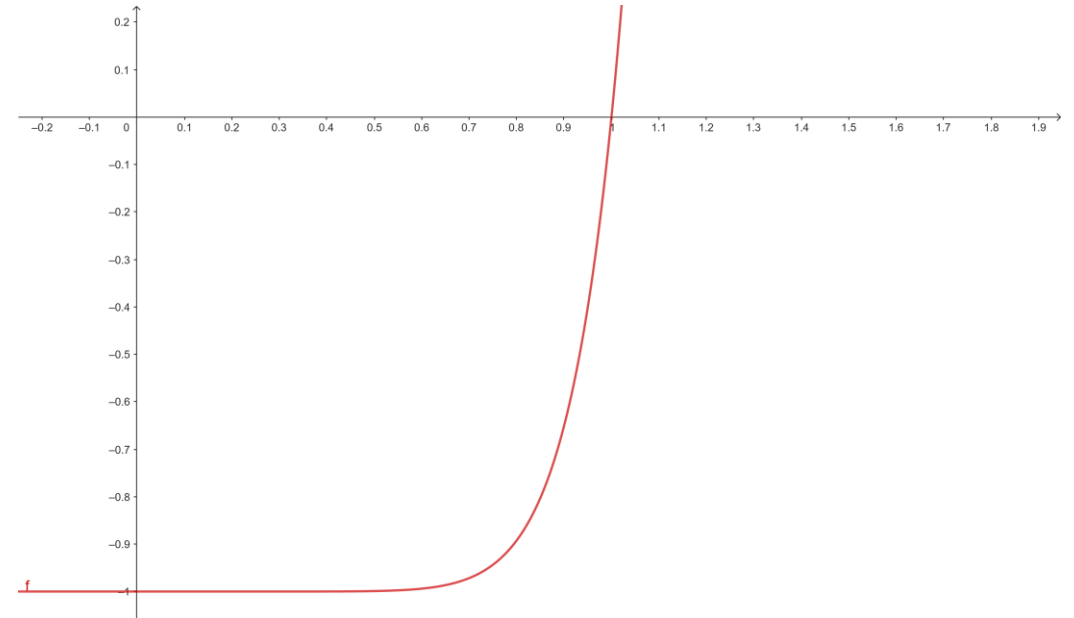


Example 2

Use bisection and false position to locate the root of

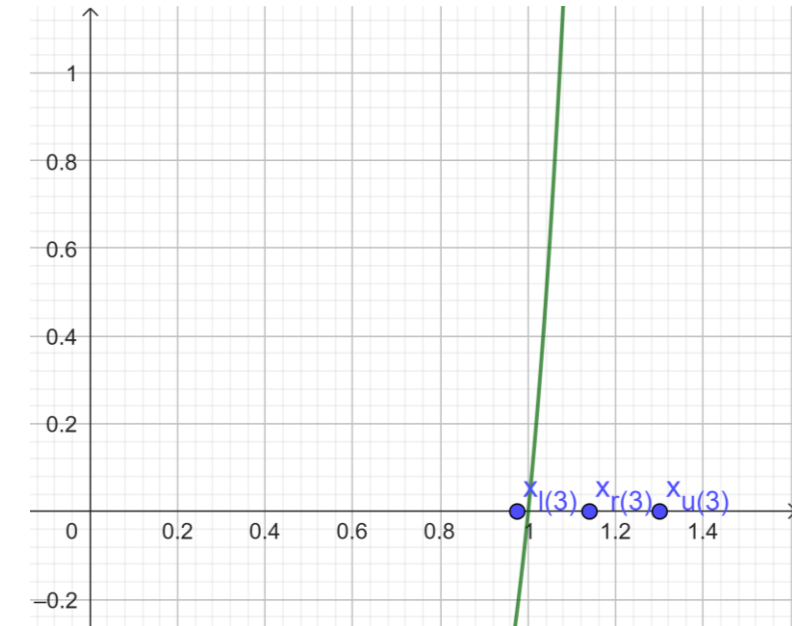
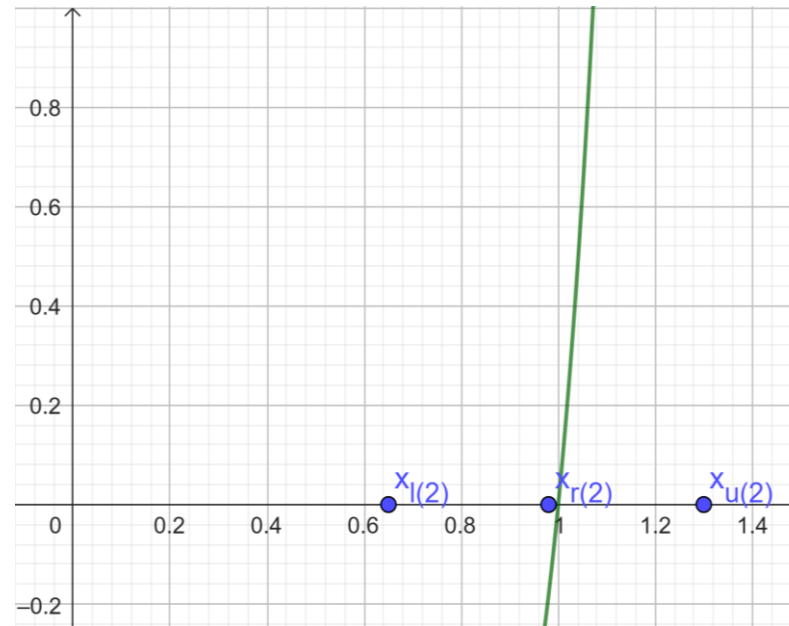
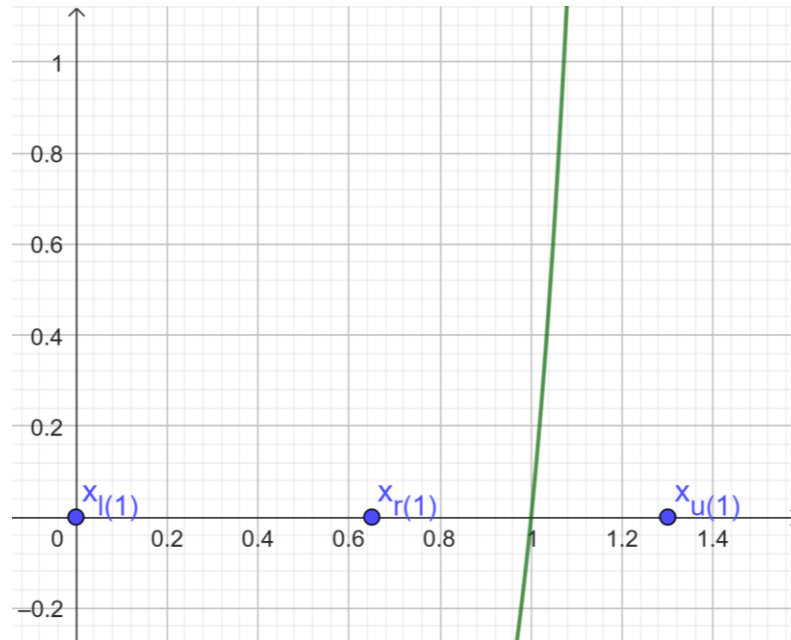
$$f(x) = x^{10} - 1$$

between $x = 0$ and 1.3 .



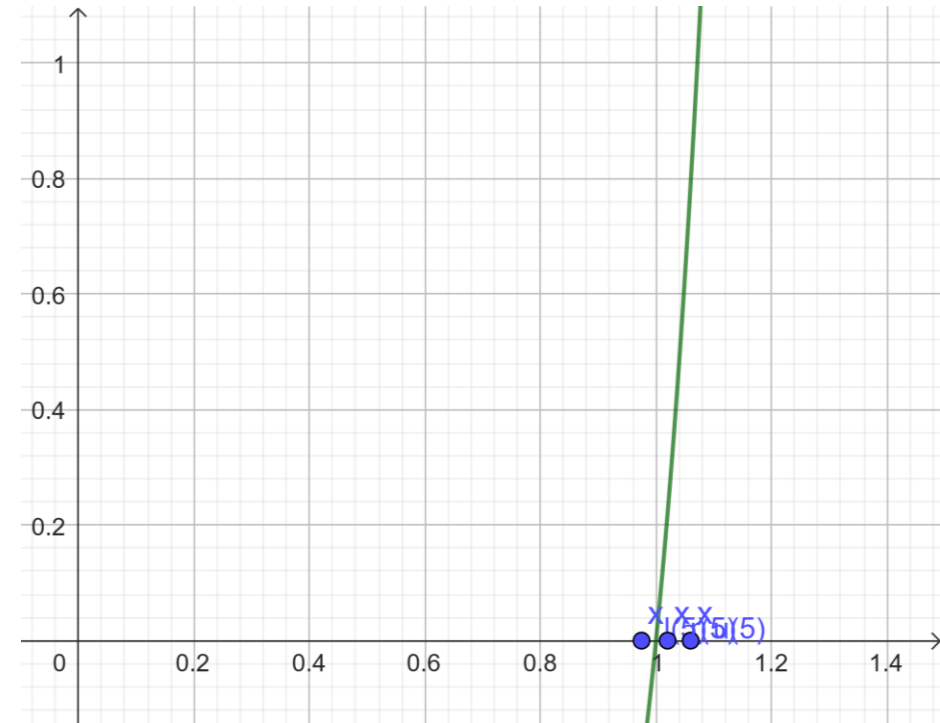
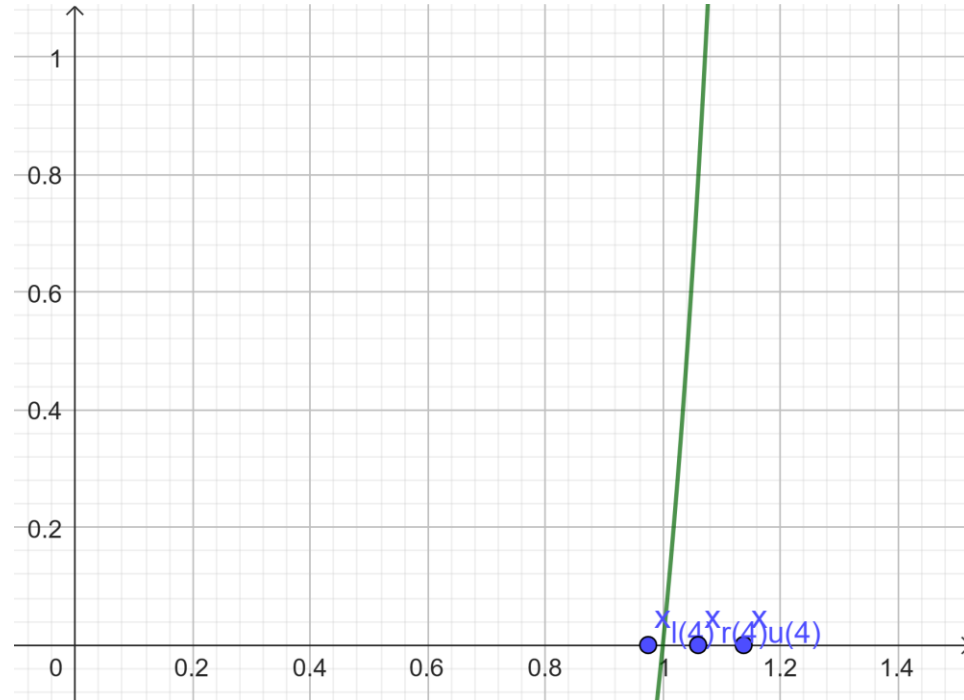


Example 2 – Bisection Method



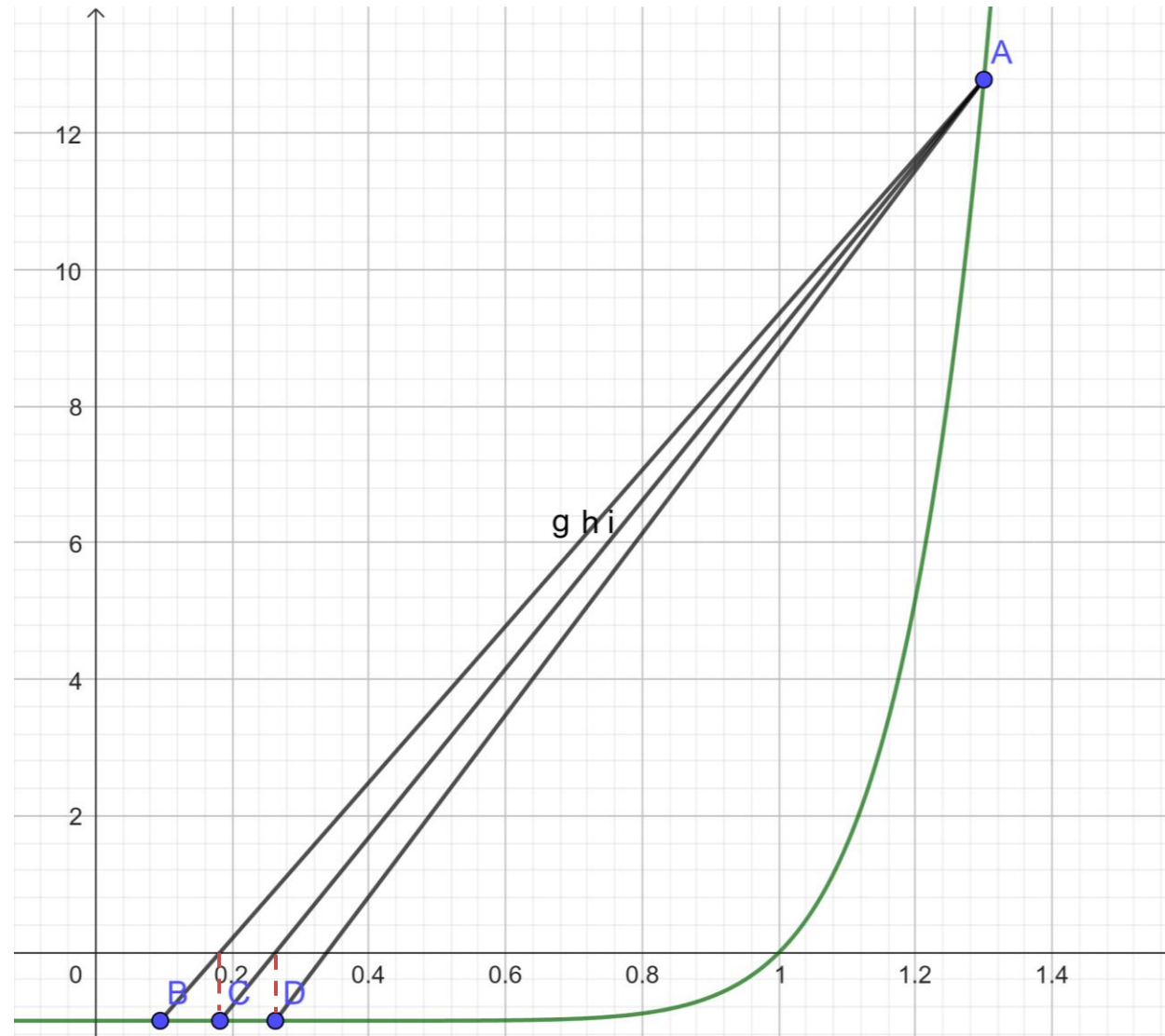


Example 2 – Bisection Method





Example 2 – False-Position Method





3. Open Methods



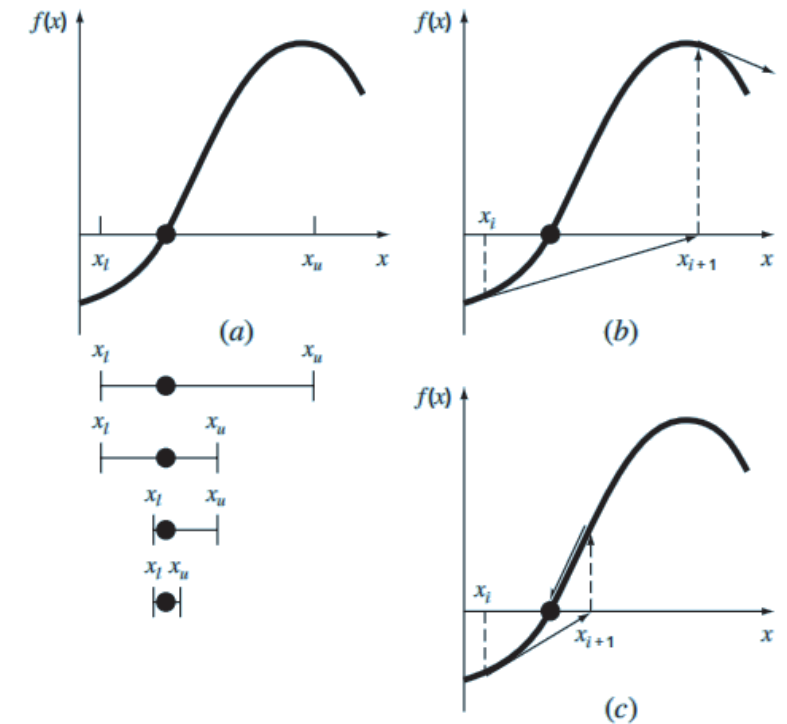
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Open vs. Bracketing Methods

- Bracketing methods (e.g., bisection) start with a lower and upper bound that contain the root and guarantee convergence by repeatedly narrowing the interval.
- Open methods rely on a formula that uses one or two initial guesses and do **not** require enclosing the root.
- Open methods can sometimes diverge (move away from the true root), although when they converge they often do so more rapidly than bracketing methods.





Simple Fixed-Point Iteration

The equation $f(x) = 0$ is rearranged into the form

$$x = g(x)$$

so that successive substitution can be applied.

This rearrangement may be done by algebraic manipulation or simply by adding x to both sides of the original equation.

$$x_{i+1} = g(x_i)$$

$$x^2 - 2x + 3 = 0$$



$$x = \frac{x^2 + 3}{2}$$

$$\sin x = 0$$



$$\sin x + x = x$$



Example 3

Use simple fixed-point iteration to locate the root of

$$0 = e^{-x} - x$$



Two-Curve (Graphical) Method

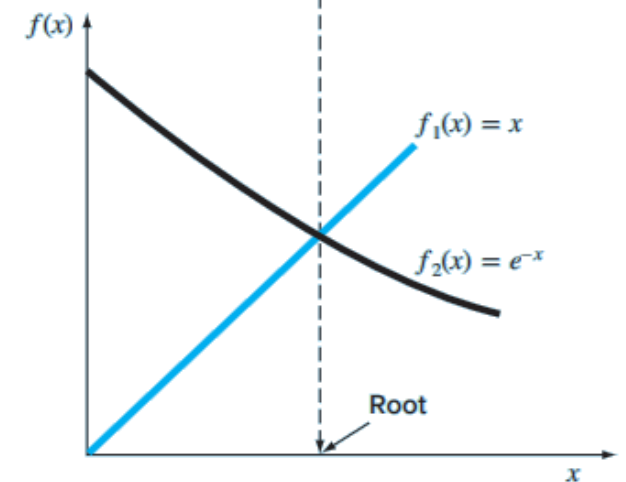
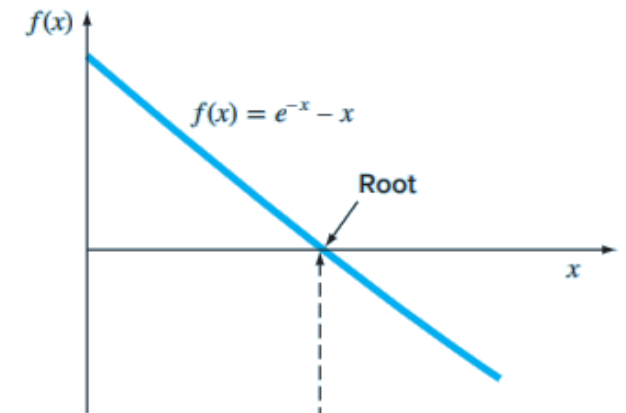
Rewrite $x = g(x)$ as two curves:

$$y_1 = x, \quad y_2 = g(x)$$

The roots of $f(x) = 0$ correspond to the intersections of these curves.

Iteration process on the plot:

1. Start at $(x_0, g(x_0))$ on the y_2 curve.
2. Move horizontally to the line $y_1 = x$, giving the new abscissa x_1 .
3. Move vertically to the curve $y_2 = g(x)$, locating $(x_1, g(x_1))$.
4. Repeat these steps to generate successive approximations $(x_2, \dots)$





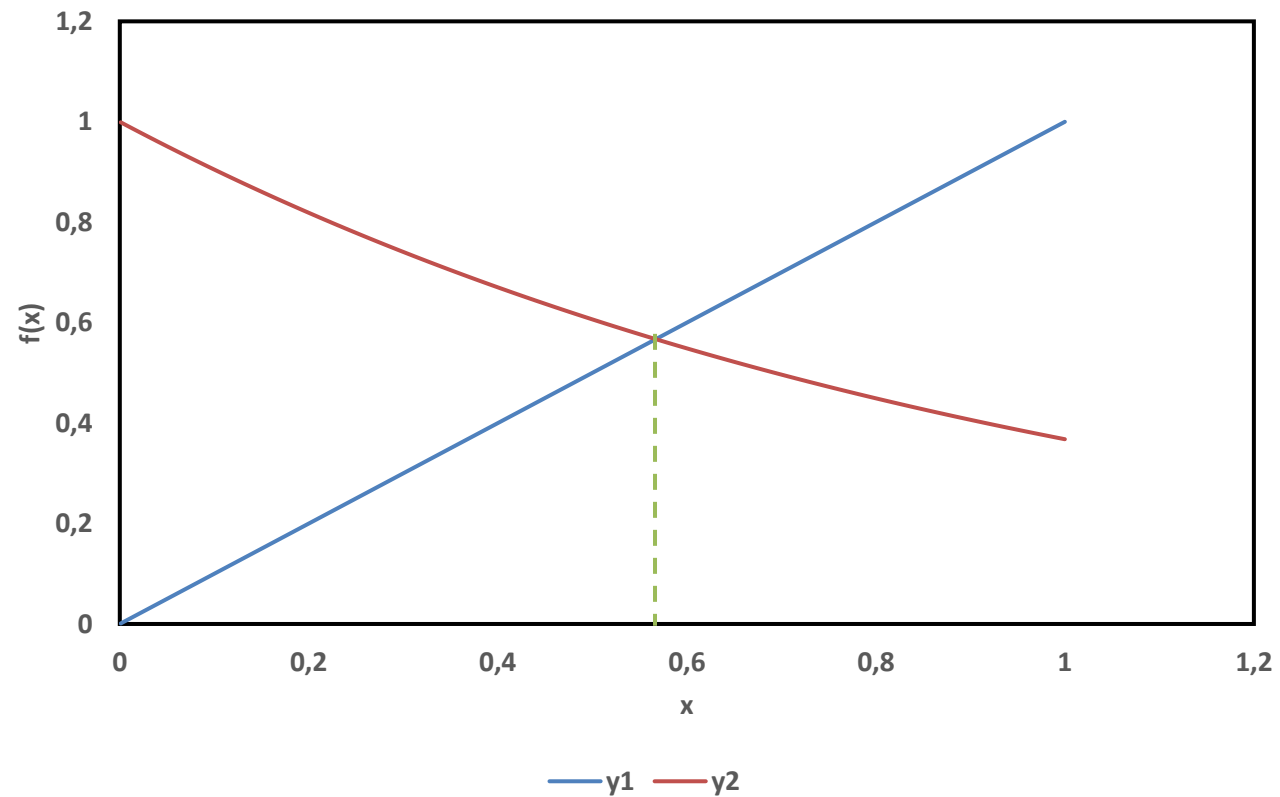
Example 4

Separate the equation into two parts and determine its root graphically.

$$0 = e^{-x} - x$$



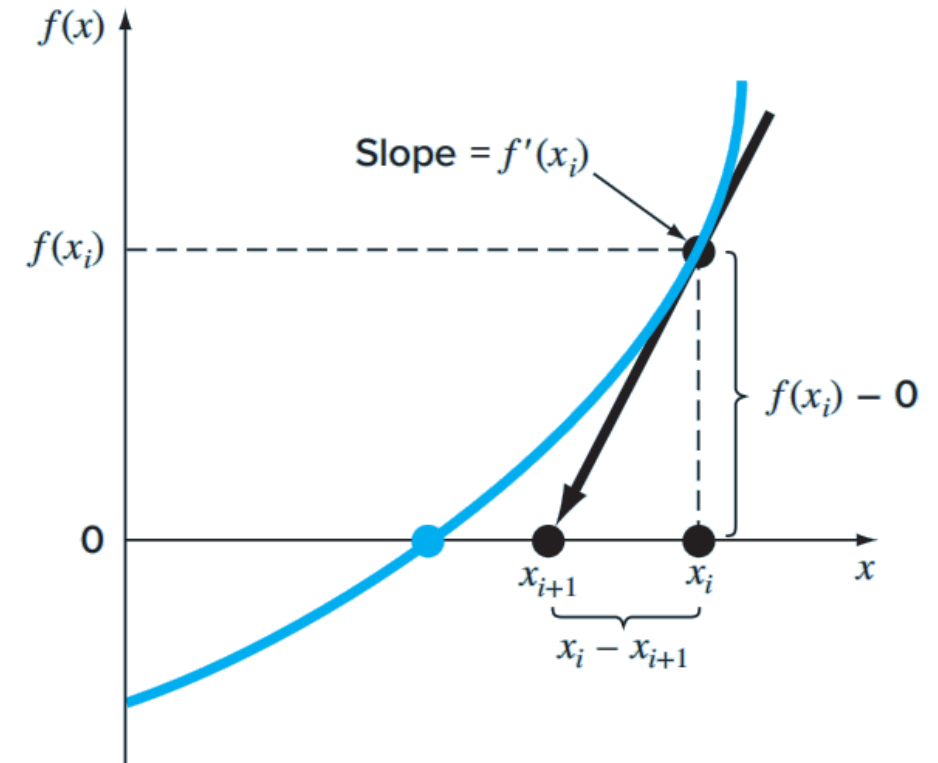
Example 4





The Newton–Raphson Method

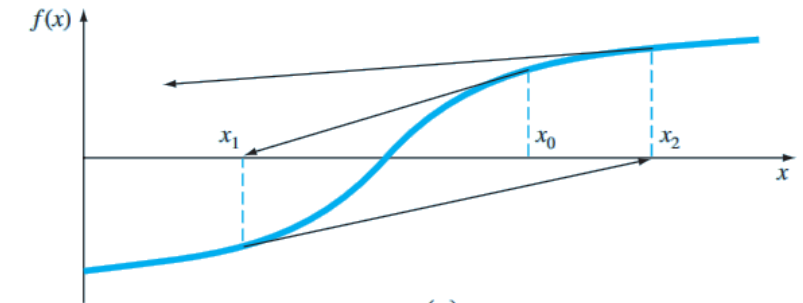
- The Newton–Raphson formula is one of the most widely used root-finding techniques.
- Geometric idea: from an initial guess x_i , draw the tangent to the curve $y = f(x)$ at $(x_i, f(x_i))$; where that tangent meets the x -axis gives a better estimate of the root.
- Starting point: $[x_i, f(x_i)]$; new approximation x_{i+1} is where the tangent crosses $y = 0$.



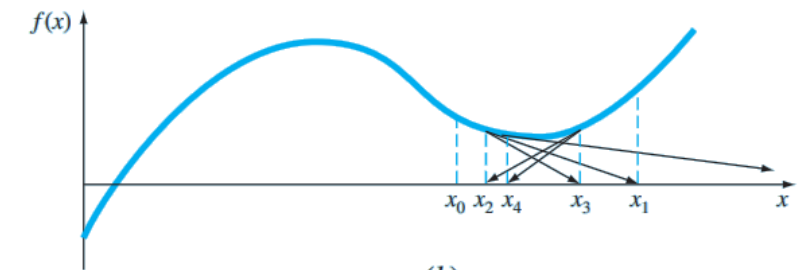


Pitfalls of the Newton–Raphson Method

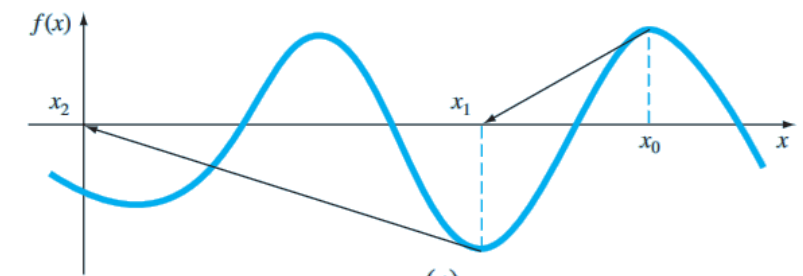
- **Inflection near root:** If $f''(x) = 0$ close to the root, successive iterates may diverge.
- **Oscillation:** The method can oscillate around a local maximum or minimum, sometimes settling at a slope near zero rather than the true root.
- **Jumping between roots:** An initial guess near one root can leap to another root farther away



(a)



(b)

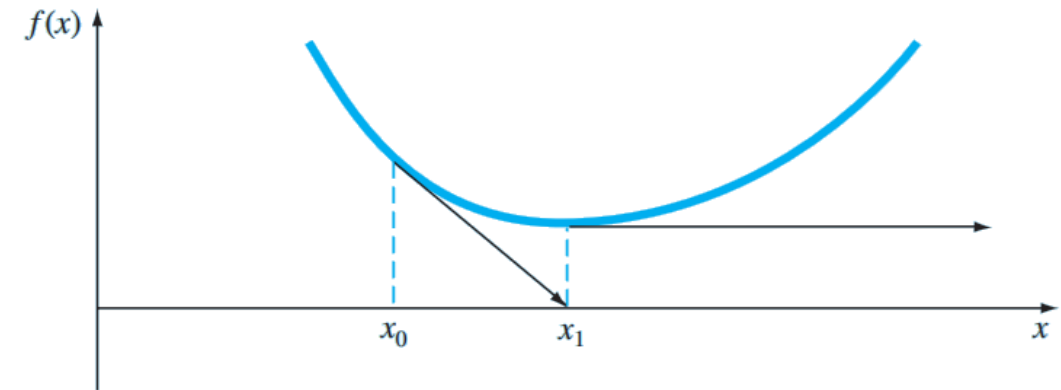


(c)



Pitfalls of the Newton–Raphson Method

- **Zero derivative:** If $f'(x) = 0$, the formula breaks down (division by zero), causing the iteration to “shoot off” horizontally and never converge.
- **No universal convergence criterion:** Success depends on the function’s behavior and closeness of the initial guess.





Algorithm for Newton–Raphson

- Implement by iterating

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

until a stopping condition is met.

- The program must be able to evaluate both $f(x)$ and its derivative $f'(x)$; typically via a user-defined function.

Algorithm for Newton–Raphson



Discipline	Real-world Applications	Typical Professions
Mechanical Engineering	Nonlinear equilibrium in FEA for stress analysis	Mechanical Design Engineer
	Gear and cam profile optimization	FEA Analyst
	Thermodynamic cycle root-finding (	Automotive Engineer
Mechatronics	Inverse kinematics for robotic arms	Mechatronic Engineer
	Servo-motor parameter calibration	Robotics Engineer
	Sensor fusion algorithm convergence	Automation Engineer
Civil Engineering	Nonlinear structural (P- Δ) stability analysis	Structural Engineer
	Slope stability and retaining-wall solutions	Geotechnical Engineer
	Settlement prediction in soil models	Civil Design Engineer
Chemical Engineering	Phase equilibrium (VLE) calculations	Process Engineer
	Reaction kinetics root-finding in reactor design	Chemical Plant Designer
	Process control loop tuning	Control Engineer
Systems Engineering	Load-flow analysis in power grids	Power Systems Engineer
	Nonlinear circuit equation solving	Control Systems Analyst
	PID-controller tuning in control systems	Simulation Engineer



Example 5

Use the Newton-Raphson method to estimate the root of $f(x) = e^{-x} - x$, employing an initial guess of $x_0 = 0$.



Secant Method

Provides a root-finding technique that avoids explicit evaluation of the derivative, useful when $f'(x)$ is difficult or costly to compute.

Derivative Approximation

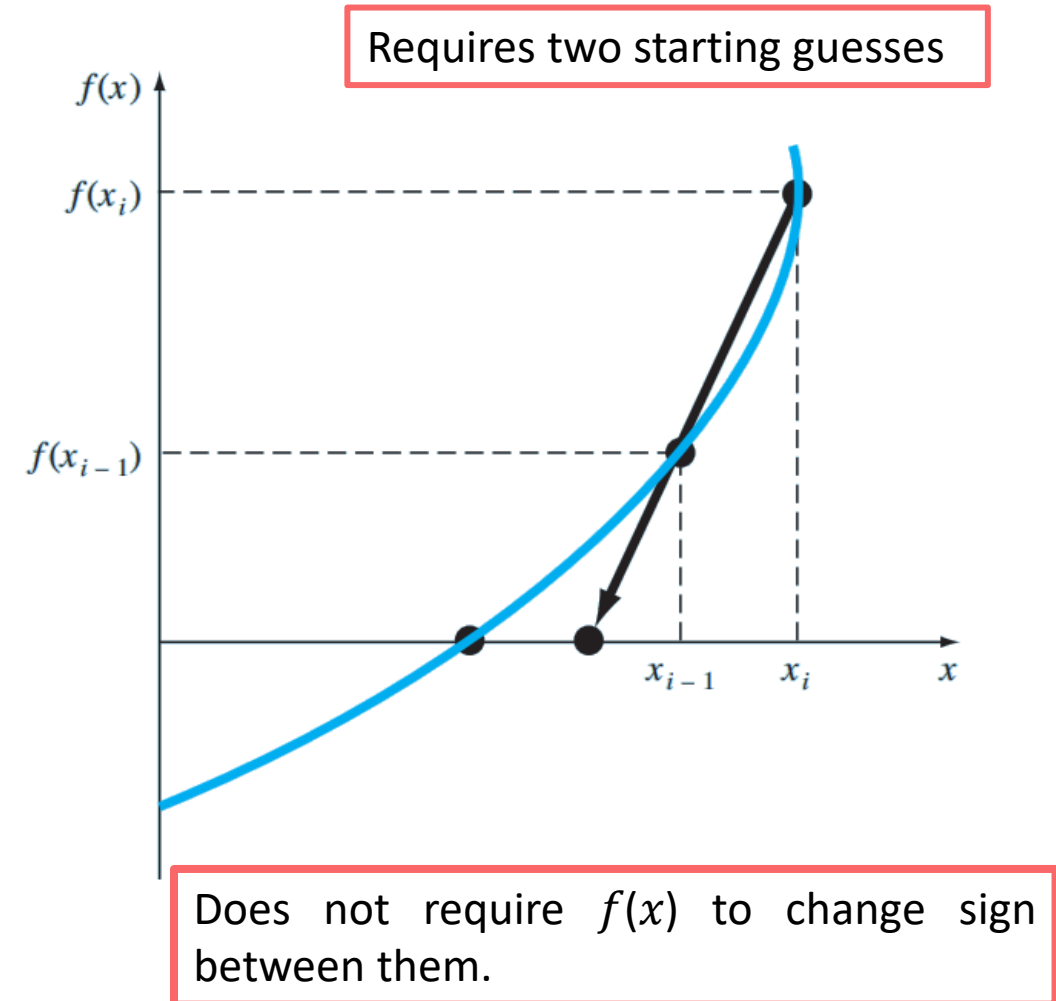
Replaces $f'(x_i)$ by a **backward finite-difference**:

$$f'(x_i) \approx \frac{f(x_{i-1}) - f(x_i)}{x_{i-1} - x_i}$$

Iterative Formula

Substituting the above into Newton's update yields:

$$x_{i+1} = x_i - \frac{f(x_i)(x_{i-1} - x_i)}{f(x_{i-1}) - f(x_i)}$$





Example 6

Use the secant method to estimate the root of $f(x) = e^{-x} - x$, employing an initial guess of $x_{i-1} = 0$ and $x_i = 1$.

$$x_{i+1} = x_i - \frac{f(x_i)(x_{i-1} - x_i)}{f(x_{i-1}) - f(x_i)}$$



Multiple Roots

Definition of multiple roots

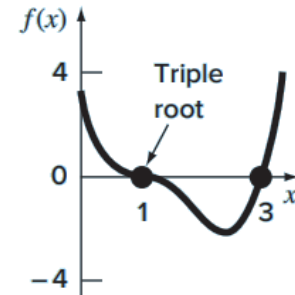
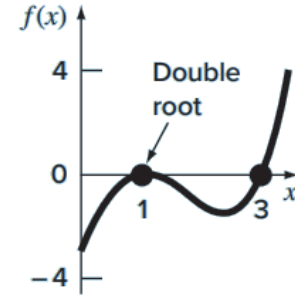
A multiple root occurs where the function is tangent to the x-axis.

Double root (m=2)

Example:

$$f(x) = (x - 3)(x - 1)^2 \Rightarrow f(x) = x^3 - 5x^2 + 7x - 3$$

Graphically, the curve touches the x-axis at $x = 1$ but does not cross it.

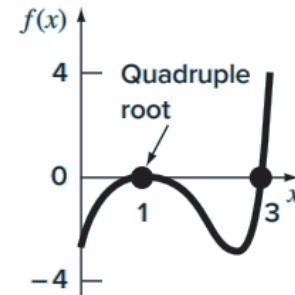


Triple root (m=3)

Example:

$$f(x) = (x - 3)(x - 1)^3 \Rightarrow f(x) = x^4 - 6x^3 + 12x^2 - 10x + 3$$

Graphically, the curve is tangent at $x=1$ and does cross the axis.





Numerical difficulties posed by multiple roots

1. Bracketing methods fail

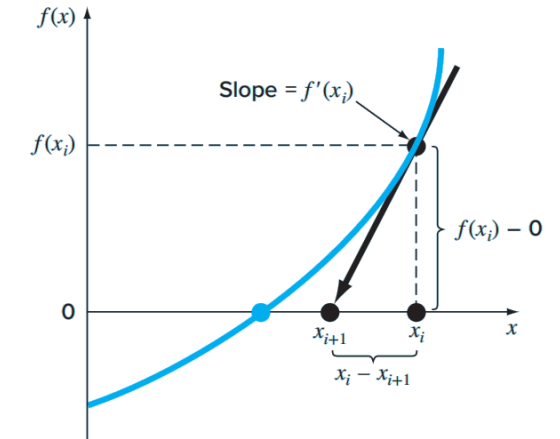
- At even roots there is no sign change, so reliable bracketing methods cannot be used.

2. Zero derivative at the root

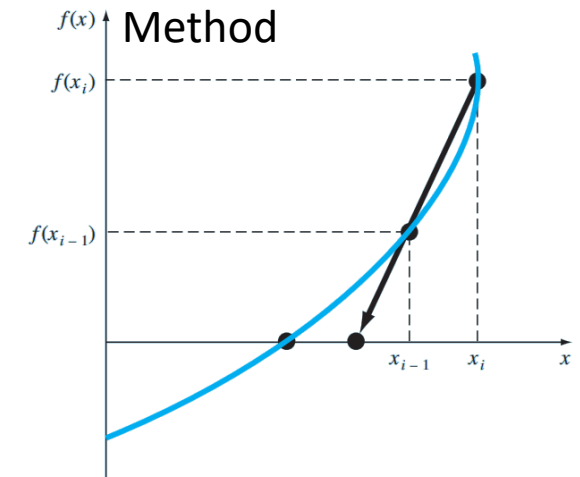
- Both $f(x)$ and $f'(x)$ go to zero, causing potential division by zero in Newton–Raphson and secant formulas.

3. Reduced convergence order

- Standard Newton–Raphson and secant methods converge only linearly (not quadratically) at multiple roots.



Newton–Raphson
Method



Secant Method



Multiplicity-independent Newton–Raphson via $u(x)$

1. Define

$$u(x) = \frac{f(x)}{f'(x)}$$

2. Then

$$x_{i+1} = x_i - \frac{u(x_i)}{u'(x_i)}$$

with

$$u'(x) = \frac{f'(x)^2 - f(x)f''(x)}{[f'(x)]^2}$$

Simplified iteration formula:

$$x_{i+1} = x_i - \frac{f(x_i)f'(x_i)}{[f'(x_i)]^2 - f(x_i)f''(x_i)}$$



Example 7

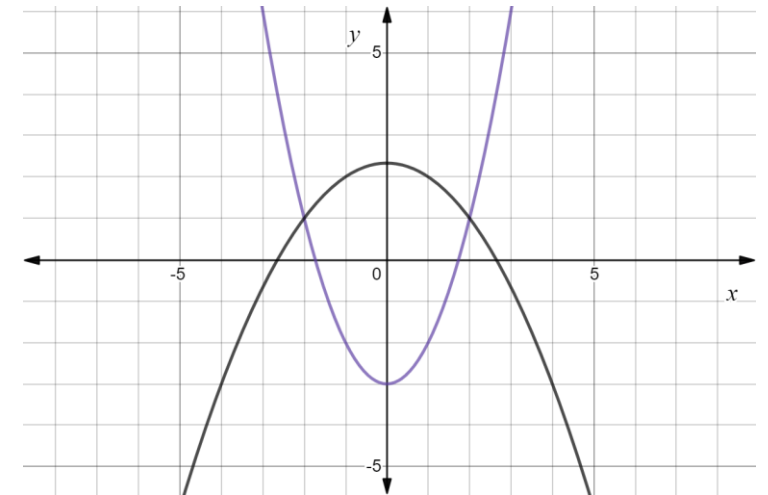
Use both the standard and modified Newton-Raphson methods to evaluate the multiple root, with an initial guess of $x_0 = 0$.

$$f(x) = x^3 - 5x^2 + 7x - 3$$



Systems of Nonlinear Equations

- A system consists of n equations $f_i(x_1, x_2, \dots, x_n) = 0$ that must be satisfied simultaneously.
- Linear systems have the form $a_1x_1 + a_2x_2 + \dots + a_nx_n - b = 0$; anything else is nonlinear.
- Example for two variables:
 - $u(x, y) = x^2 + xy - 10 = 0$
 - $v(x, y) = y + 3xy^2 - 57 = 0$





Multi-Equation Newton–Raphson

- Use a first-order multivariable Taylor series for each function:

$$u_{i+1} = u_i + (x_{i+1} - x_i) \frac{\partial u_i}{\partial x} + (y_{i+1} - y_i) \frac{\partial u_i}{\partial y}$$

$$v_{i+1} = v_i + (x_{i+1} - x_i) \frac{\partial v_i}{\partial x} + (y_{i+1} - y_i) \frac{\partial v_i}{\partial y} \quad \Rightarrow$$

$$x_{i+1} = x_i - \frac{u_i \frac{\partial v_i}{\partial y} - v_i \frac{\partial u_i}{\partial y}}{\frac{\partial u_i}{\partial x} \frac{\partial v_i}{\partial y} - \frac{\partial u_i}{\partial y} \frac{\partial v_i}{\partial x}}$$

$$y_{i+1} = y_i - \frac{v_i \frac{\partial u_i}{\partial x} - u_i \frac{\partial v_i}{\partial x}}{\frac{\partial u_i}{\partial x} \frac{\partial v_i}{\partial y} - \frac{\partial u_i}{\partial y} \frac{\partial v_i}{\partial x}}$$

- Setting $u_{i+1} = v_{i+1} = 0$ yields two linear equations for x_{i+1} and y_{i+1}
- Solving via Cramer's rule gives the update formulas:



Example 8

- Use the multiequation Newton-Raphson method to determine roots of the equations. Note that a correct pair of roots is $x = 2$ and $y = 3$. Initiate the computation with guesses of $x = 1.5$ and $y = 3.5$.

$$u(x, y) = x^2 + xy - 10 = 0$$

$$v(x, y) = y + 3xy^2 - 57 = 0$$

$$x_{i+1} = x_i - \frac{u_i \frac{\partial v_i}{\partial y} - v_i \frac{\partial u_i}{\partial y}}{\frac{\partial u_i}{\partial x} \frac{\partial v_i}{\partial y} - \frac{\partial u_i}{\partial y} \frac{\partial v_i}{\partial x}}$$

$$y_{i+1} = y_i - \frac{v_i \frac{\partial u_i}{\partial x} - u_i \frac{\partial v_i}{\partial x}}{\frac{\partial u_i}{\partial x} \frac{\partial v_i}{\partial y} - \frac{\partial u_i}{\partial y} \frac{\partial v_i}{\partial x}}$$



4. Roots of Polynomials



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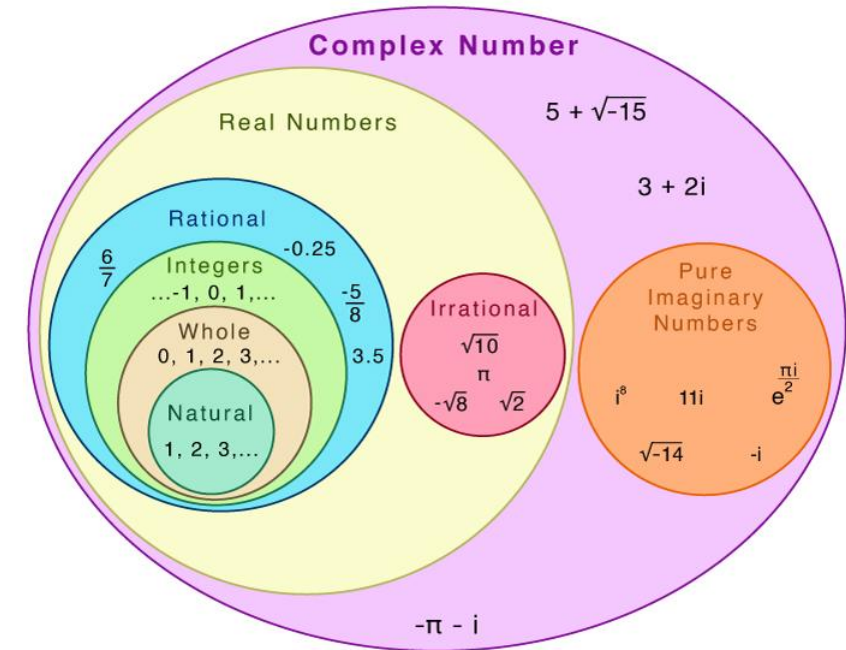




Introduction

- The general form of an n^{th} -order polynomial:

$$f_n(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$$
- Coefficients a_i may be complex, but here we assume they are real.
- A polynomial of order n has exactly n roots (real or complex), not necessarily distinct.
- If n is odd, there is at least one real root.





Need for Numerical Methods

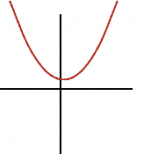
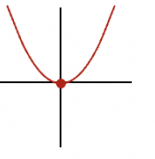
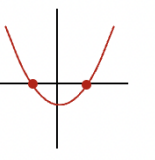
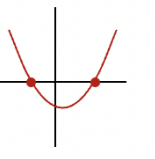
- Quadratic formulas solve second-order polynomials analytically.
- Higher-order polynomials lack general closed-form solutions.
- Numerical root-finding methods are required for efficiency and generality.



Conventional Methods

Real vs. Complex Roots

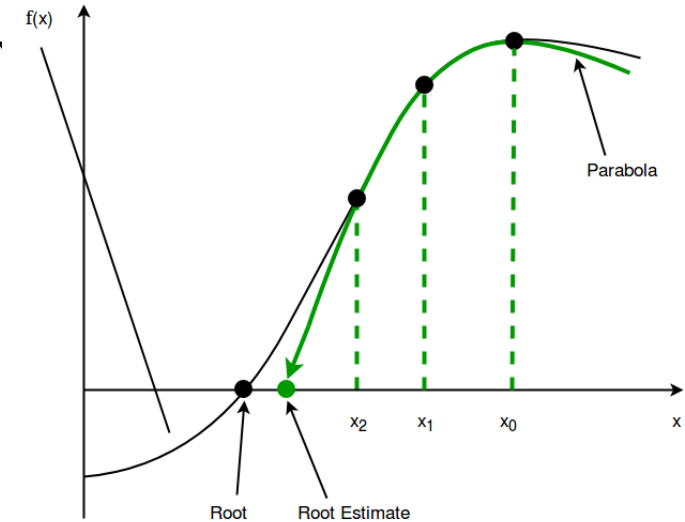
- If only real roots exist, both bracketing and open methods can work, but:
 - Bracketing requires good interval guesses.
 - Open methods (e.g. Newton–Raphson) can diverge without a suitable starting point.
- If complex roots may occur, **bracketing fails** (no sign-change criterion for complex numbers).

Value of Discriminant	Number/Nature of Roots	Graph
$b^2 - 4ac < 0$	2 Imaginary Solutions	
$b^2 - 4ac = 0$	1 Real & Rational Solution	
$b^2 - 4ac > 0$ (perfect square)	2 Real & Rational Solutions	
$b^2 - 4ac > 0$ (non-perfect square)	2 Real & Irrational Solutions	



Conventional Methods

- **Newton–Raphson**
 - A viable open-method: supports complex arithmetic (with deflation) to locate real and complex roots.
 - However, it is **susceptible to convergence problems**.
- **Specialized Polynomial Solvers**
 - To overcome these issues, one related open-methods are presented next: **Müller's Method**





Müller's Method

- Secant method fits a straight line through two points and projects to the x-axis.
- Müller's method fits a parabola through three points for root estimation.

Parabolic Model

- Define the interpolating parabola as

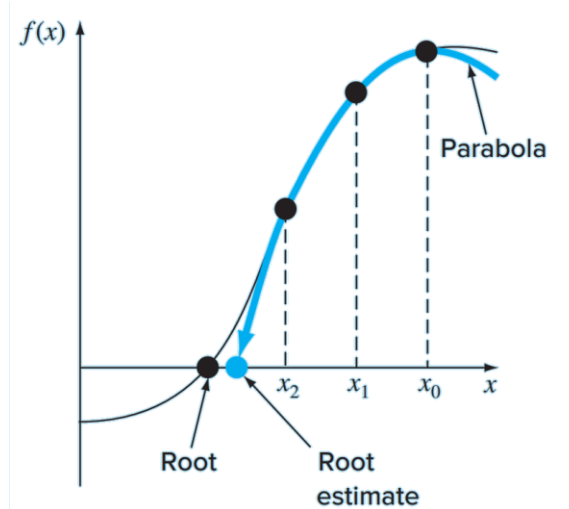
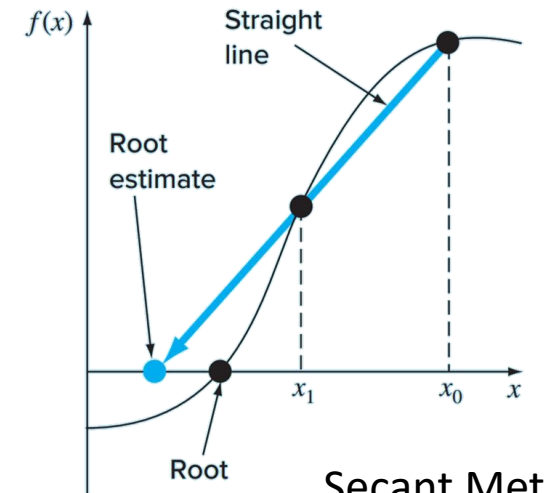
$$f_2(x) = a(x - x_2)^2 + b(x - x_2) + c$$

- The parabola must satisfy

$$f(x_0) = a(x_0 - x_2)^2 + b(x_0 - x_2) + c$$

$$f(x_1) = a(x_1 - x_2)^2 + b(x_1 - x_2) + c$$

$$f(x_2) = a(x_2 - x_2)^2 + b(x_2 - x_2) + c$$





Müller's Method

Determination of Coefficients

- From the third equation, directly obtain

$$c = f(x_2)$$

- Substitute into the first two to get two equations in a and b :

$$\begin{aligned} f(x_0) - f(x_2) &= a(x_0 - x_2)^2 + b(x_0 - x_2) \\ f(x_1) - f(x_2) &= a(x_1 - x_2)^2 + b(x_1 - x_2) \end{aligned}$$

Finite-Difference Notation

- Let

$$\begin{aligned} h_0 &= x_1 - x_0, & h_1 &= x_2 - x_1, \\ \delta_0 &= \frac{f(x_1) - f(x_0)}{x_1 - x_0}, & \delta_1 &= \frac{f(x_2) - f(x_1)}{x_2 - x_1} \end{aligned}$$

Formulas for a and b

$$\begin{aligned} a &= \frac{\delta_1 - \delta_0}{h_1 + h_0} \\ b &= a h_1 + \delta_1 \\ c &= f(x_2) \end{aligned}$$



Müller's Method

Root Estimate via Quadratic Formula

- Use the numerically stable form:

$$x_3 = x_2 + \frac{-2c}{b \pm \sqrt{b^2 - 4ac}}.$$

- Choose the sign in the denominator that matches the sign of b , to maximize the denominator and yield the estimate closest to x_2 .

Iteration and Point Replacement

- **Strategy 1 (Real roots only):** Keep the two points nearest to the new estimate x_3 .
- **Strategy 2 (Real and complex roots):** Shift points so that (x_1, x_2, x_3) replace (x_0, x_1, x_2) for the next cycle.



Example 9

Apply the Müller's Method using the starting estimates $x_0 = 4.5$, $x_1 = 5.5$, and $x_2 = 5.0$ to locate one real solution of the cubic polynomial

$$f(x) = x^3 - 13x - 12$$

(For reference, the exact roots of this equation are -3 , -1 and 4 .)



5. Root Location with Software Packages



Root Location with Software Packages

- Software packages offer powerful tools for finding roots of functions.
- This section introduces some of the most useful ones.

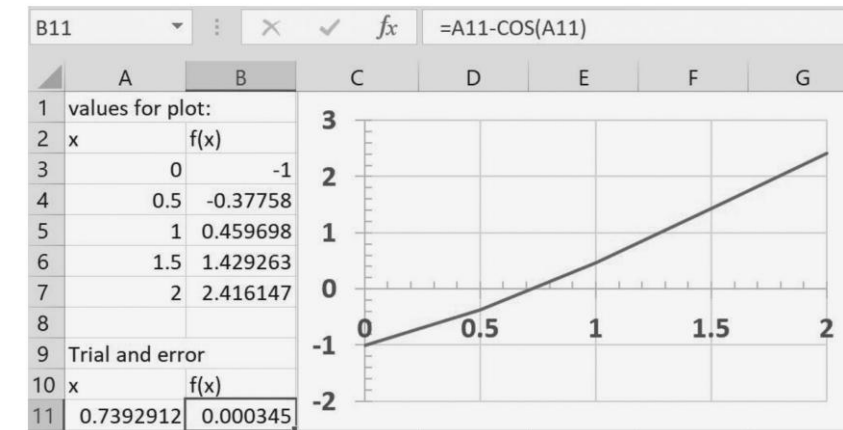




Excel

- A spreadsheet like Excel can locate a root by trial-and-error:

1. Enter a value for x in one cell.
2. Compute $f(x)$ in another cell using that x .
3. Adjust the x cell until the $f(x)$ cell approaches zero.
4. Use Excel's plotting capabilities to obtain a good initial guess.





Excel

Excel also provides two built-in root-finding tools:

- **Goal Seek**
 - Varies a single parameter to drive an equation to a specified value (in this case, zero).
- **Solver**
 - Systematically adjusts one or more variables to satisfy constraints and optimize a target cell.





Example 10

Employ Goal Seek to determine the root of the transcendental function:

$$f(x) = x - \cos x$$



Example 11

Remember that in example 8 we obtained the solution of the following set of simultaneous equations

- $u(x, y) = x^2 + xy - 10 = 0$
- $v(x, y) = y + 3xy^2 - 57 = 0$

Note that a correct pair of roots is $x = 2$ and $y = 3$. Use Solver to determine the roots using initial guesses of $x = 1$ and $y = 3.5$.



MATLAB Root-Finding (fzero)

- MATLAB can locate roots of both algebraic and transcendental equations.
- It is especially powerful for manipulating and finding the roots of polynomials.
- The basic syntax for finding one root of a single function is:
$$\text{root} = \text{fzero}(\text{f}, \text{x0}, \text{options})$$
 - **f**: handle to the function whose root you seek
 - **x0**: initial guess (or two guesses to bracket the root)
 - **options**: optimization parameters (set via optimset; defaults used if omitted)
- If you supply two guesses, they are assumed to bracket a single root.



MATLAB Root-Finding (fzero)

Function	Description
fzero	Root of single function.
roots	Find polynomial roots.
poly	Construct polynomial with specified roots.
polyval	Evaluate polynomial.
polyvalm	Evaluate polynomial with matrix argument.
residue	Partial-fraction expansion (residues).
polyder	Differentiate polynomial.
conv	Multiply polynomials.
deconv	Divide polynomials.



Example 12

Use the MATLAB function `fzero` to find the roots of

$$f(x) = x^{10} - 1$$

within the interval $x_l = 0$ and $x_u = 4$.



Solution - Example 12

Defining the Function Handle

```
f = @(x) x.^10 - 1;
```

- Uses an anonymous function $@(x)$ to represent $f(x) = x^{10} - 1$.
- The elementwise power `.^` ensures compatibility if x were ever a vector.
- This handle f can now be passed directly to MATLAB solvers like `fzero`.

Setting the Search Interval

```
xl = 0; % lower bound  
xu = 4; % upper bound
```

- x_l and x_u define the bracket $[x_l, x_u]$ where we expect a sign change in f .
- A valid sign change over this interval guarantees – by the Intermediate Value Theorem – at least one real root inside.



Solution - Example 12

Checking the Sign Change

```
fprintf('f(%.1f) = %.2f (debe ser negativo)\n', xl, f(xl));  
fprintf('f(%.1f) = %.2f (debe ser positivo)\n', xu, f(xu));
```

- Evaluates and prints $f(x_l)$ and $f(x_u)$ to verify that $f(x_l) < 0$ and $f(x_u) > 0$.
- These diagnostic messages reassure you that `fzero` will have a valid initial bracket.

Calling `fzero` to Find the Root

```
raiz = fzero(f, [xl xu]);
```

- `fzero` locates a point $raiz$ in $[x_l, x_u]$ where $f(raiz) = 0$.
- Internally, it combines bisection with a secant method for fast and reliable convergence.



Solution - Example 12

Displaying the Root with High Precision

```
fprintf('\nLa raíz de x^10 - 1 en [%.1f, %.1f] es:\n', xl, xu);  
fprintf('  x = %.8f\n', raiz);
```

- Prints a human-readable summary of the result, showing the interval and the found root.
- The format specifier `%.8f` ensures eight decimal places, highlighting the numerical accuracy.

Verifying the Solution

```
valor_f = f(raiz);
```

```
fprintf('Verificación: f(%.8f) = %.2e (cercano a cero)\n', raiz, valor_f);
```

- Re-evaluates f at the computed root and prints the outcome in scientific notation (`%.2e`).
- Confirms that the residual $f(\text{raiz})$ is on the order of 10^{-12} or smaller, validating that the solver indeed found a true root.



6. Case Studies: Application of Engineering



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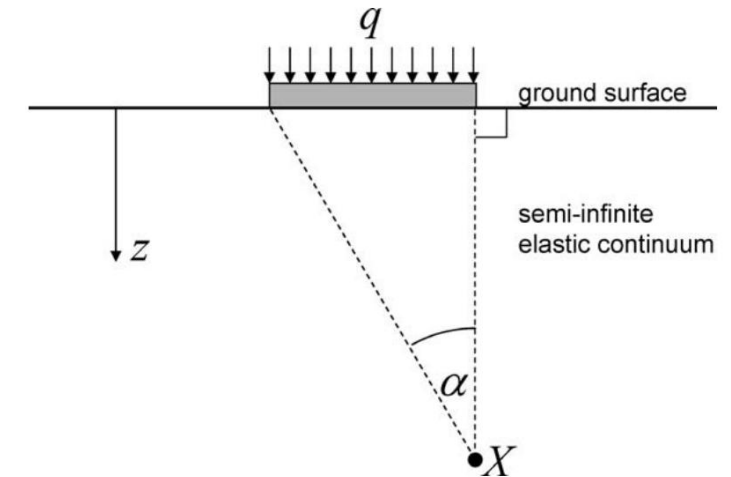


Civil Engineering – Example 13

Consider the elastic half-space, in which a strip footing applies a uniform pressure q at the ground surface. The vertical stress σ_z at point X (located at depth z and subtending an angle α at the edge of the footing) is given by Boussinesq's formula:

$$\sigma_z = \frac{q}{\pi} (\alpha + \sin \alpha \cos \alpha)$$

Use the Newton–Raphson Method to estimate the value of α (in radians) that makes the vertical stress equal to 25 % of the footing pressure.





Chemical Engineering – Example 14

Compute the molar volume of CO₂ and O₂ using the Ideal Gas Law and the van der Waals equation.

Parameters:

- $R = 0.082054 \text{ L} \cdot \text{atm}/(\text{mol} \cdot \text{K})$
- CO₂: $a = 3.592, b = 0.04267$
- O₂: $a = 1.360, b = 0.03183$

$$\left[P + a \left(\frac{n}{V} \right)^2 \right] \left(\frac{V}{n} - b \right) = RT$$

Conditions:

- Temperatures $T = 300, 500, 700 \text{ K}$
- Pressures $p = 1, 10, 100 \text{ atm}$

Solve the van der Waals equation with the Newton–Raphson method.



7. Summary



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Summary

Trade-offs in root-finding methods

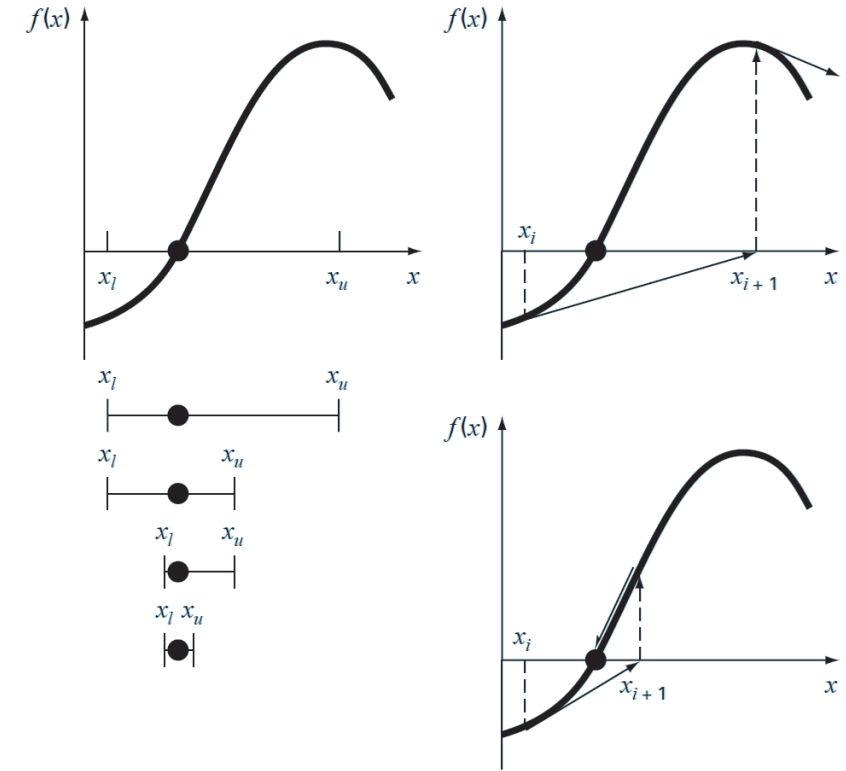
- Graphical methods offer insight into function behavior (e.g., multiple roots) but are time-consuming and imprecise for final values.
- Bracketing methods (e.g., bisection, false-position) guarantee convergence by maintaining an interval where the sign of $f(x)$ changes, but converge relatively slowly.
- Open methods (e.g., fixed-point iteration, Newton–Raphson, secant) often converge faster, yet can diverge if the initial guess or function behavior is unfavorable.



Summary

Bracketing vs. open techniques

- **Bracketing:** requires two guesses that enclose the root; always converges, but rate is at best linear.
- **Open:** uses one (or two non-enclosing) guesses to extrapolate new estimates; convergence can be superlinear or quadratic, yet is conditional on a good starting point.





Summary

Newton–Raphson and its variants

- **Newton–Raphson:** quadratic convergence when the derivative is known analytically; major drawback is the need for an explicit derivative.
- **Secant method:** finite-difference approximation of the derivative—slower initial convergence but avoids analytic differentiation and accelerates as the estimate improves.
- **Modified Newton methods:** can target multiple roots simultaneously but require expressions for higher derivatives.



Summary

Method	Type	Guesses	Convergence	Stability	Programming	Comments
Direct	Analytical	—	—	—	—	Imprecise
Graphical	Visual	—	—	—	—	
Bisection	Bracketing	2	Slow	Always	Easy	
False-position	Bracketing	2	Slow/medium	Always	Easy	
Modified FP	Bracketing	2	Medium	Always	Easy	
Fixed-point iteration	Open	1	Slow	Possibly divergent	Easy	Requires evaluation of $f'(x)$
Newton-Raphson	Open	1	Fast	Possibly divergent	Easy	
Modified Newton-Raphson	Open	1	Fast (multiple), medium (single)	Possibly divergent	Easy	
Secant	Open	2	Medium/fast	Possibly divergent	Easy	Requires evaluation of $f'(x)$ and $f''(x)$
Modified secant	Open	1	Medium/fast	Possibly divergent	Easy	Initial guesses do not have to bracket the root
Brent	Hybrid	1 or 2	Medium	Always (for 2 guesses)	Moderate	
Müller	Polynomials	2	Medium/fast	Possibly divergent	Moderate	Robust
Bairstow	Polynomials	2	Fast	Possibly divergent	Moderate	



Summary

Method	Formulation	Graphical Interpretation	Errors and Stopping Criteria
Bisection	$x_r = \frac{x_l + x_u}{2}$ If $f(x_l)f(x_r) < 0$, $x_u = x_r$ $f(x_l)f(x_r) > 0$, $x_l = x_r$	Bracketing methods:	Stopping criterion: $\left \frac{x_r^{\text{new}} - x_r^{\text{old}}}{x_r^{\text{new}}} \right 100\% \leq \epsilon_s$
False position	$x_r = x_u - \frac{f(x_u)(x_l - x_u)}{f(x_l) - f(x_u)}$ If $f(x_l)f(x_r) < 0$, $x_u = x_r$ $f(x_l)f(x_r) > 0$, $x_l = x_r$		Stopping criterion: $\left \frac{x_r^{\text{new}} - x_r^{\text{old}}}{x_r^{\text{new}}} \right 100\% \leq \epsilon_s$
Newton-Raphson	$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$		Stopping criterion: $\left \frac{x_{i+1} - x_i}{x_{i+1}} \right 100\% \leq \epsilon_s$ Error: $E_{i+1} = O(E_i^2)$
Secant	$x_{i+1} = x_i - \frac{f(x_i)(x_{i-1} - x_i)}{f(x_{i-1}) - f(x_i)}$		Stopping criterion: $\left \frac{x_{i+1} - x_i}{x_{i+1}} \right 100\% \leq \epsilon_s$